Harness² Engineering

Authors: Ph.D N.Momtselidze with AI enjineering

“Harness² Engineering” isn’t a standard mathematical term; the exponent is a metaphor for using engineering to improve both a system and the harness that tests and guides it.

Let:

1. = the system’s capability
2. = the quality of its harness: tests, tools, constraints, and feedback A useful analogy is:

[= C H]

A capable system with a weak harness can behave unreliably. A strong harness helps it work consistently and reveals where it needs improvement.

The “squared” idea is the feedback loop:

[H_{n+1} = H_n +]

As the harness improves, it helps improve the system; as the system changes, it reveals how the harness should improve. In that sense, Harness² Engineering means engineering the product and engineering the process that makes the product dependable.